

F-4. Transition Pathway of Teak Tree				
Forest / Shrub Land Conservation	High	Medium		
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions									
Measures	The different phases of growth, development, and ecological interactions that a teak tree (<i>Tectona grandis</i>) goes through from germination to maturity, as well as its function in the ecosystem over the course of its life cycle, are referred to as the transition pathway. Environmental variables, land management techniques, and particular forestry objectives can all have an impact on this pathway. The transition pathway of a teak tree (<i>Tectona grandis</i>) in Bangladesh's CHT region includes a variety of essential measures meant to promote the tree's growth and ensure sustainable forest management. Because of the region's diverse geography, soil types, and climate, growing teak presents both opportunities and challenges requiring for careful planning and intervention at every step of the tree's lifecycle. The teak tree transition pathway in CHT combines site-specific management practices, such as sustainable harvesting, pest control, soil conservation, and careful selection, with the goal of maximizing growth and guaranteeing that teak continues to be a valuable, renewable resource that benefits the local ecosystem.								
	Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77
Applicable Sites	Mono culture teak gardens								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
			√		√		√		
Design options and performance	• Selective Clearing of Invasive Species – Reduces competition and allows native trees to flourish. • Natural Seed Dispersal Enhancement – Encourages seedlings from native trees to regenerate. • Soil Protection & Water Retention Measures – Uses mulching, contour bunds, and vegetation strips to enhance soil fertility.								
Feasibility criteria	• Flat terrain and well-drained, fertile soils with a high organic matter content are favorable for teak growth. Because of the high topography and variety of soil types found in the CHT, it is important to carefully choose locations where the soil can sustain teak growth while overcoming challenges such soil erosion and planting difficulties. • In order to reduce degradation and promote teak development, effective soil conservation practices (such as terracing and mulching) are crucial given the steep slopes and erosion vulnerability of the CHT. Long-term sustainability requires taking into consideration teak's environmental benefits in terms of								

	carbon storage. Plantations of mixed species could be taken into consideration to preserve biodiversity. Long-term sustainability of teak plantations, including their ability to renew naturally or with the aid of assisted natural regeneration (ANR) techniques, requires careful management. Measures for conserving forests and reforestation are essential. Furthermore, teak plantations ought to be developed to resist climate extremes like pests, droughts, and variations in rainfall patterns.			
Operation and maintenance	The region's challenging landscape and environmental conditions require frequent upkeep of the teak tree transition pathway. First and foremost, successful planting requires selecting high-quality teak seeds and properly preparing the planting location. Healthy growth is ensured by regular weeding, pest control, and soil conservation measures. Replanting and natural regeneration, along with sustainable harvesting methods, assure the plantation's long-term survival.			
Cost	High			
Benefits				
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			

	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	- Teak trees reduce the risk of erosion caused by heavy rainfall by stabilizing the soil with their deep root systems, particularly on the steep slopes of the Hill Tracts. By increasing circulation of nutrients and storing organic matter from fallen leaves and branches, it may also improve the quality of the soil. This sustains soil health over time and aids in the restoration of degraded soils. - Although teak is a monoculture, properly run teak plantations can contribute to ecological diversity in the area as long as they are maintained to encourage the process of regeneration or mixed with other native species. The water content of the surrounding environment is likewise maintained by their canopy. The plants support the stability of the ecosystem by providing native species a place to live. - Teak trees, which grow quickly, enable the sequestration carbon, which reduces the adverse effects of climate change. By offering shade, regulating local temperatures, and averting drastic changes in the surrounding environment, the trees aid in the regulation of microclimates. Teak farming supports sustainable land-use practices by providing a financially feasible alternative for shifting cultivation or uncontrolled deforestation.			
Remarks	This measure will aid landscape restoration in CHT of Bangladesh along with creating alternative livelihood opportunities.			